1-amazon-sales-analysis

August 23, 2024

[1]:
```
#Importing Liabraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
sns.set_style('whitegrid')
```

[2]:
```
#Extracting CSV Dataset from System using Pandas Liabrary
data = pd.read_csv(r"C:\Users\sarth\Documents\DS_f proj\1_Amazon Sales data.
↪csv")
```

[3]:
```
#Dataset Head Information
data.head()
```

[3]:
```
                            Region                Country        Item Type  \
0             Australia and Oceania                 Tuvalu        Baby Food
1  Central America and the Caribbean                Grenada           Cereal
2                            Europe                 Russia  Office Supplies
3                Sub-Saharan Africa  Sao Tome and Principe           Fruits
4                Sub-Saharan Africa                 Rwanda  Office Supplies

  Sales Channel Order Priority Order Date   Order ID  Ship Date  Units Sold  \
0       Offline              H  5/28/2010  669165933  6/27/2010        9925
1        Online              C  8/22/2012  963881480  9/15/2012        2804
2       Offline              L   5/2/2014  341417157   5/8/2014        1779
3        Online              C  6/20/2014  514321792   7/5/2014        8102
4       Offline              L   2/1/2013  115456712   2/6/2013        5062

   Unit Price  Unit Cost  Total Revenue  Total Cost  Total Profit
0      255.28     159.42     2533654.00  1582243.50     951410.50
1      205.70     117.11      576782.80   328376.44     248406.36
2      651.21     524.96     1158502.59   933903.84     224598.75
3        9.33       6.92       75591.66    56065.84      19525.82
4      651.21     524.96     3296425.02  2657347.52     639077.50
```

[4]:
```
#All columns in a dataset
data.columns
```

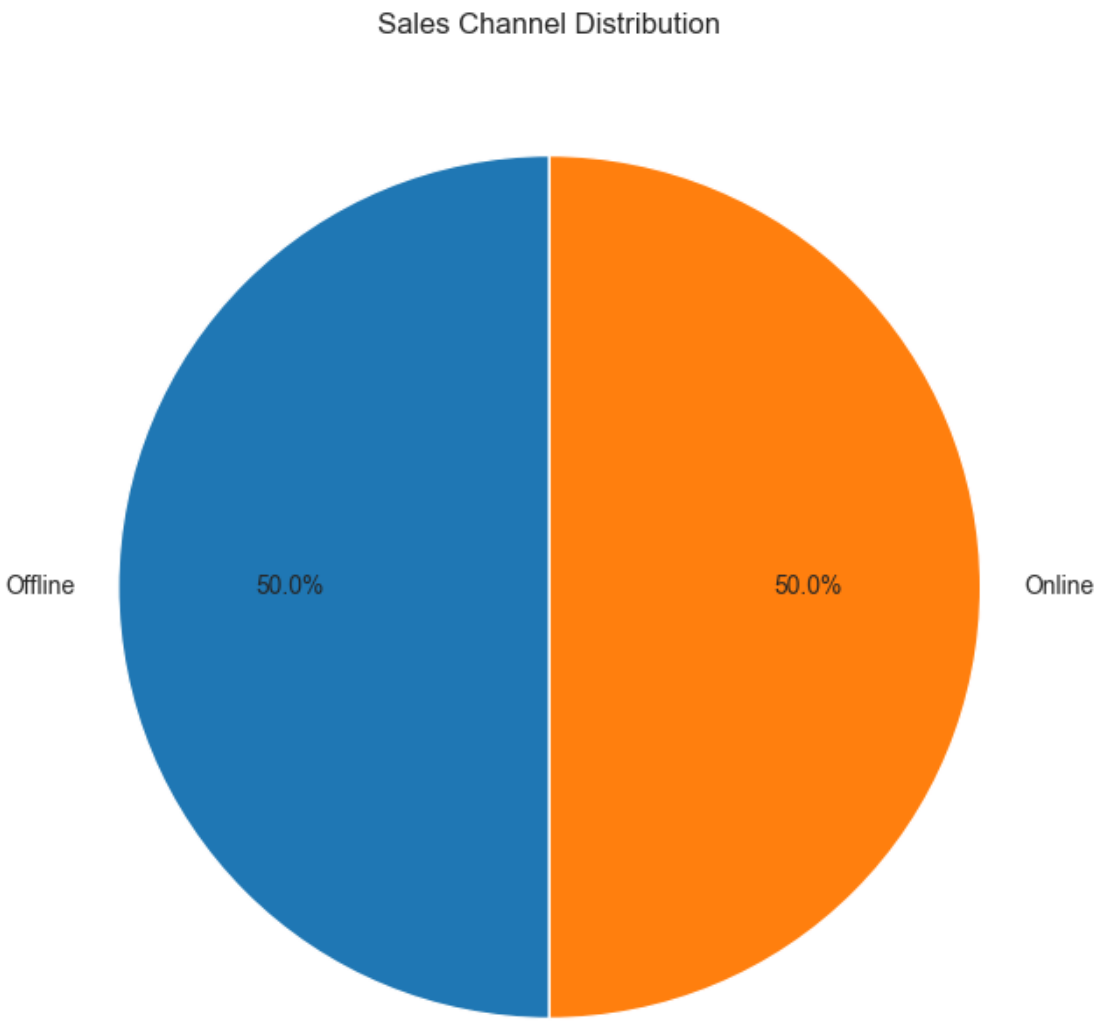
[4]:
```
Index(['Region', 'Country', 'Item Type', 'Sales Channel', 'Order Priority',
       'Order Date', 'Order ID', 'Ship Date', 'Units Sold', 'Unit Price',
       'Unit Cost', 'Total Revenue', 'Total Cost', 'Total Profit'],
      dtype='object')
```

[5]:
```
#Checking NULL values
data.isnull().sum()
```

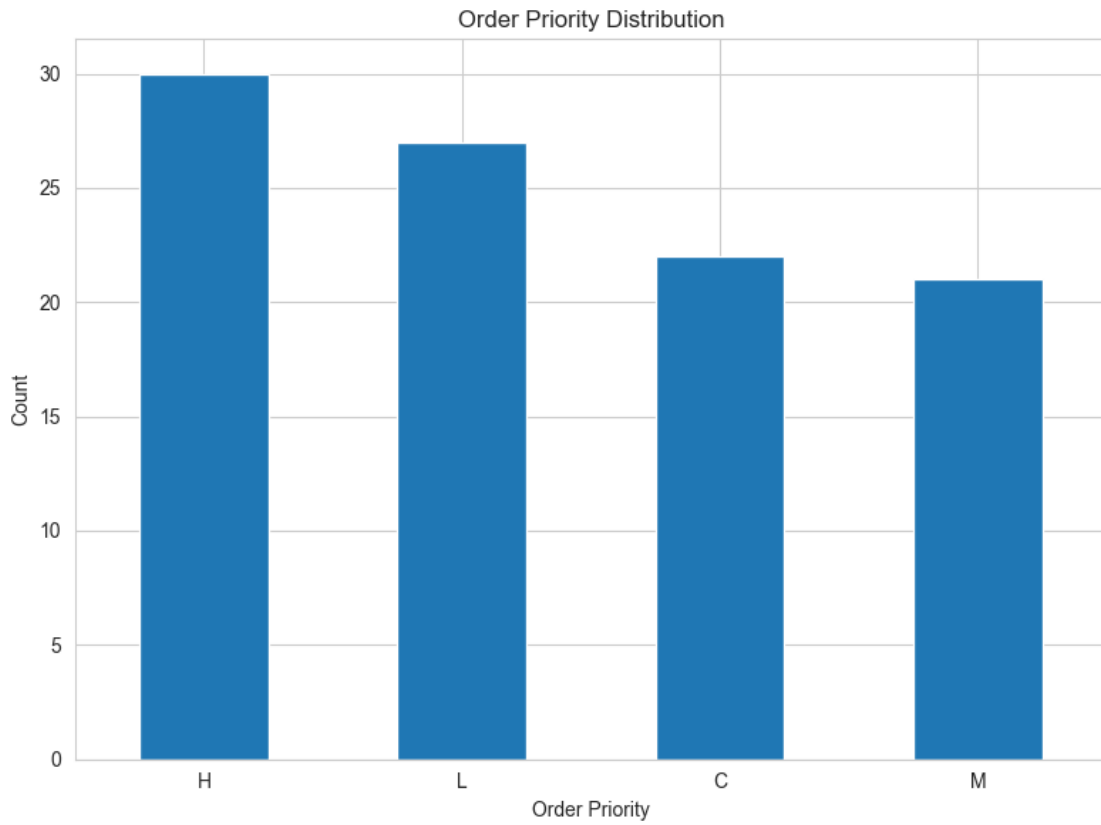
[5]:
```
Region            0
Country           0
Item Type         0
Sales Channel     0
Order Priority    0
Order Date        0
Order ID          0
Ship Date         0
Units Sold        0
Unit Price        0
Unit Cost         0
Total Revenue     0
Total Cost        0
Total Profit      0
dtype: int64
```

1 There are no missing values in dataset.

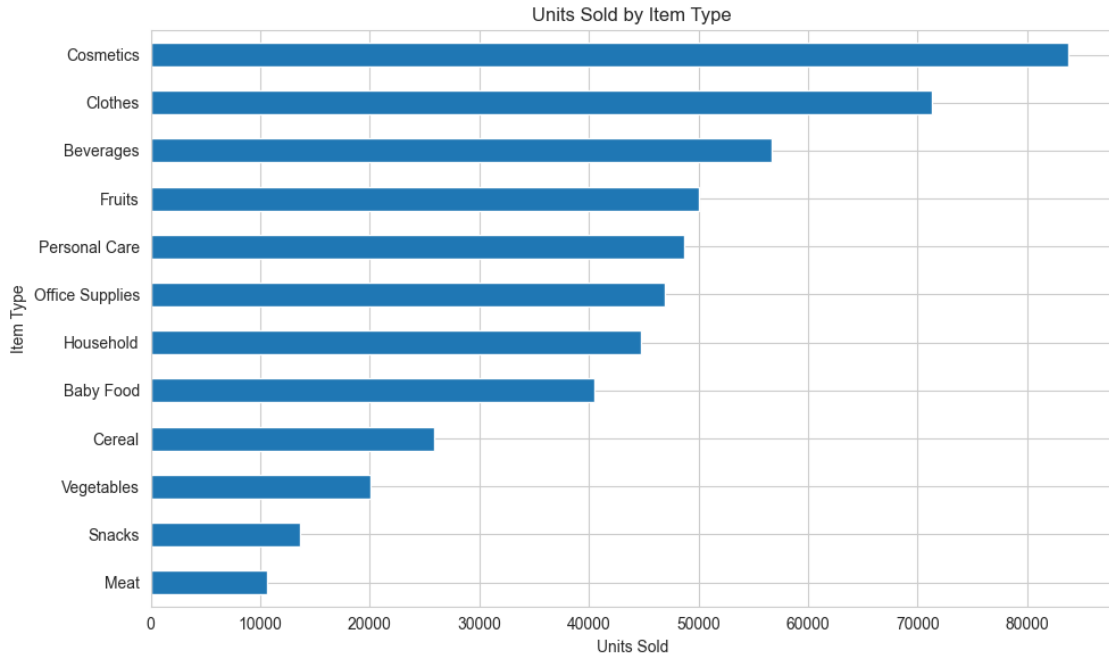
[6]:
```
# Calculate the total sales for each sales channel using pie chart.
sales_channel_distribution = data['Sales Channel'].value_counts()

# Create a pie chart
plt.figure(figsize=(8, 8))
plt.pie(sales_channel_distribution, labels=sales_channel_distribution.index,␣
 ↪autopct='%1.1f%%', startangle=90)
plt.title('Sales Channel Distribution')
plt.show()
```



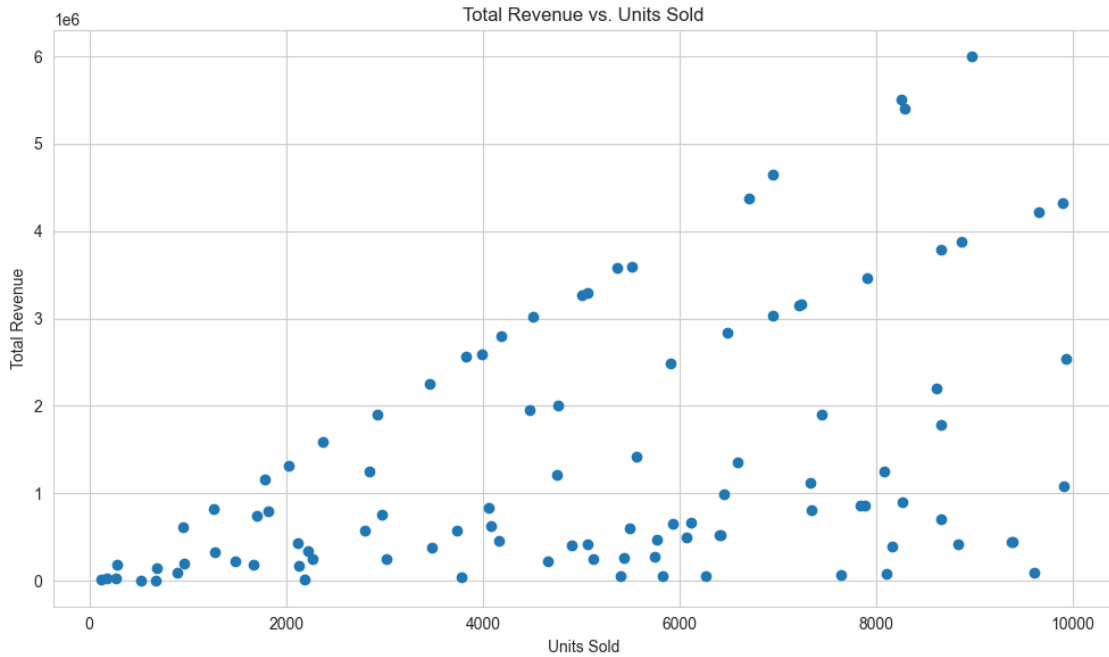
[7]:
```
# Group data by order priority and count occurrences using stacked bar chart.
order_priority_distribution = data['Order Priority'].value_counts()

# Create a stacked bar chart
plt.figure(figsize=(8, 6))
order_priority_distribution.plot(kind='bar', stacked=True)
plt.xlabel('Order Priority')
plt.ylabel('Count')
plt.title('Order Priority Distribution')
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



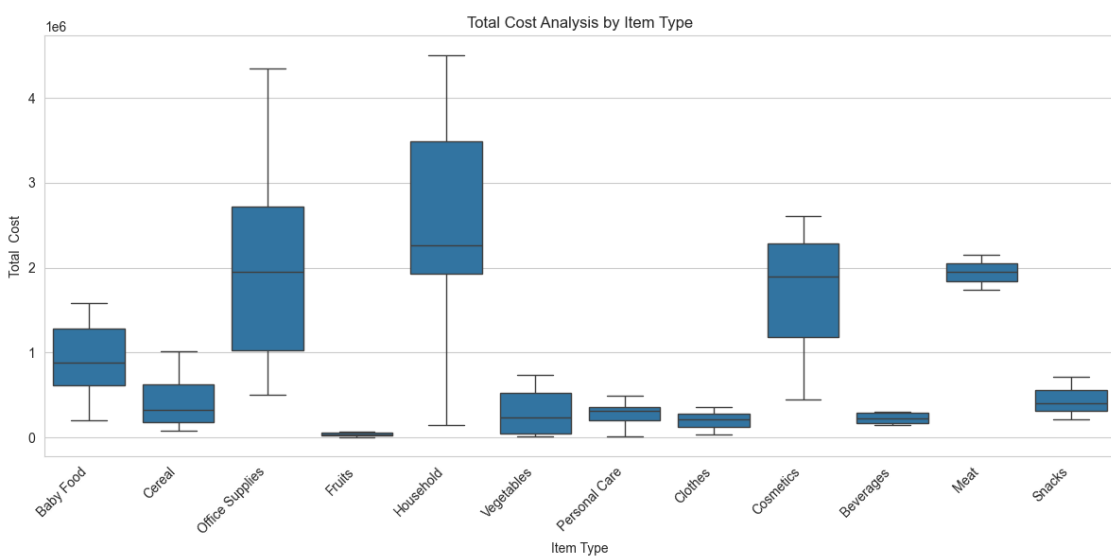
[8]:
```
# Group data by item type and calculate total units sold by bar plot.
item_type_sales = data.groupby('Item Type')['Units Sold'].sum().
 ↪sort_values(ascending=True)

# Create a horizontal bar chart
plt.figure(figsize=(10, 6))
item_type_sales.plot(kind='barh')
plt.ylabel('Item Type')
plt.xlabel('Units Sold')
plt.title('Units Sold by Item Type')
plt.tight_layout()
plt.show()
```



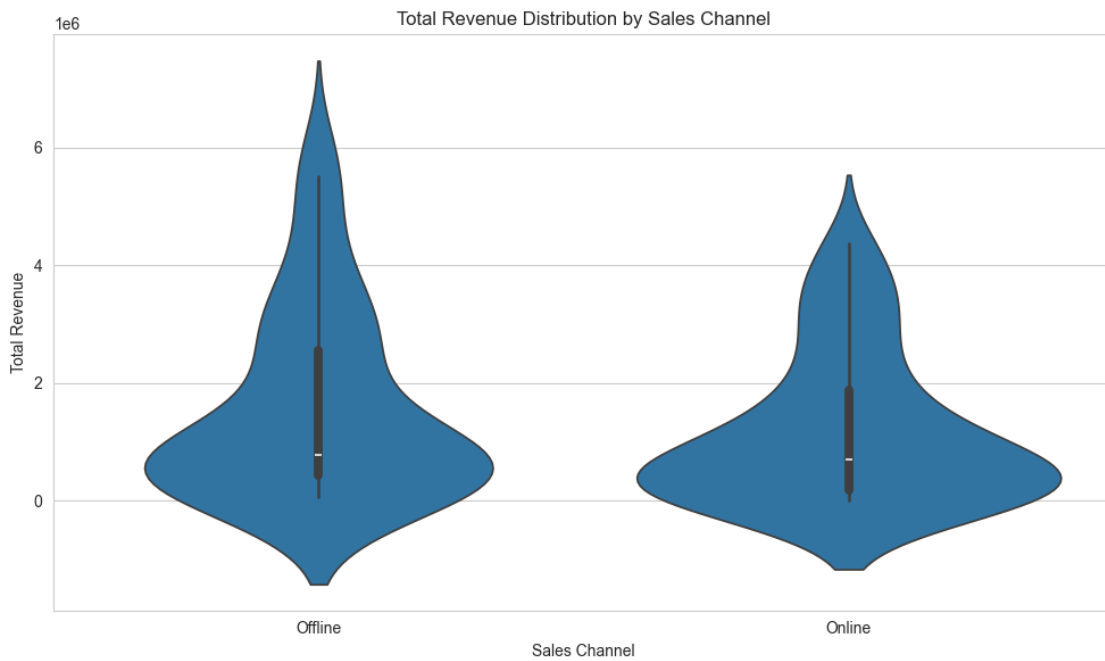
[9]:
```
# Total Revenue vs units sold by scatter plot.
plt.figure(figsize=(10, 6))
plt.scatter(data['Units Sold'], data['Total Revenue'])
plt.xlabel('Units Sold')
plt.ylabel('Total Revenue')
plt.title('Total Revenue vs. Units Sold')
plt.grid(True)
plt.tight_layout()
plt.show()
```



[10]:
```
# Analysis of cost analysis by item type by box plot.
plt.figure(figsize=(12, 6))
sns.boxplot(x='Item Type', y='Total Cost', data=data)
plt.xlabel('Item Type')
plt.ylabel('Total  Cost')
plt.title('Total Cost Analysis by Item Type')
plt.xticks(rotation=45, ha='right')  # Rotate x-axis labels for readability
plt.tight_layout()
plt.show()
```



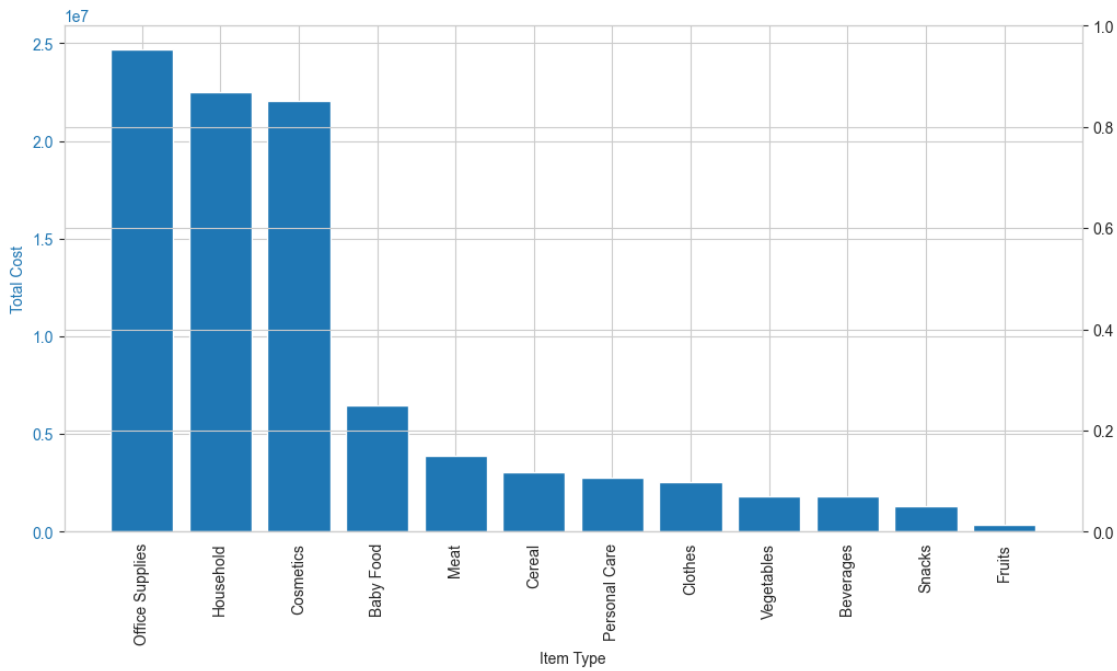
[11]:
```
# Analysis of revenue distribution by sales channel using violin plot.
plt.figure(figsize=(10, 6))
sns.violinplot(x='Sales Channel', y='Total Revenue', data=data)
plt.xlabel('Sales Channel')
plt.ylabel('Total Revenue')
plt.title('Total Revenue Distribution by Sales Channel')
plt.tight_layout()
plt.show()
```



[12]:
```
# Group data by product and calculate total sale using pareto chart (␣
↪cumulative bar chart)
product_sales = data.groupby('Item Type')['Total Cost'].sum().
↪sort_values(ascending=False)

# Calculate cumulative sales and percentage
cumulative_sales = product_sales.cumsum()
total_sales = cumulative_sales.iloc[-1]
cumulative_percentage = (cumulative_sales / total_sales) * 100

# Create a Pareto chart (cumulative bar chart)
fig, ax1 = plt.subplots(figsize=(12, 6))

# Bar chart for sales
```

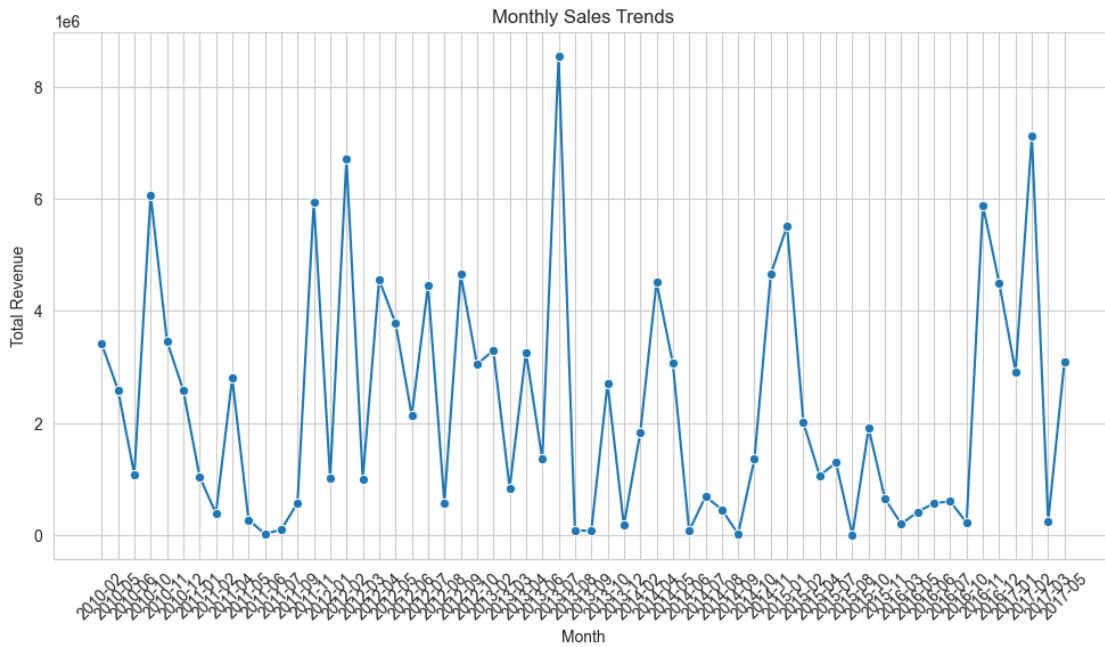
```
ax1.bar(product_sales.index, product_sales.values, color='tab:blue')
ax1.set_xlabel('Item Type')
ax1.set_ylabel('Total Cost', color='tab:blue')
ax1.tick_params(axis='y', labelcolor='tab:blue')
plt.xticks(rotation=90)

# Line chart for cumulative percentage
ax2 = ax1.twinx()
ax2
```

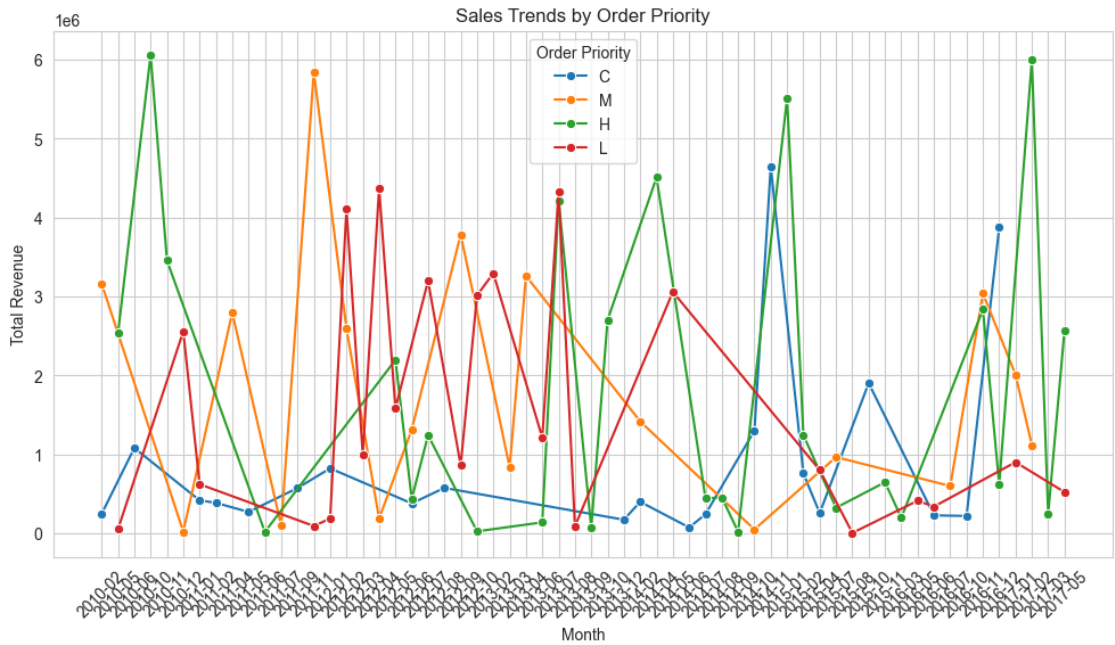
[12]: <Axes: >



[13]:
```
# Monthly Sales Trend using line chart.
data['Order Date'] = pd.to_datetime(data['Order Date'], errors='coerce')

# Extract month and year for grouping
data['Month'] = data['Order Date'].dt.to_period('M')

# Group by month and sum the total revenue
monthly_sales = data.groupby('Month')['Total Revenue'].sum().reset_index()

# Convert 'Month' column to string for plotting # This is the change
monthly_sales['Month'] = monthly_sales['Month'].astype(str)

# Plot the monthly sales trends
```

```
plt.figure(figsize=(12, 6))
sns.lineplot(x='Month', y='Total Revenue', data=monthly_sales, marker='o')
plt.xticks(rotation=45)
plt.title('Monthly Sales Trends')
plt.xlabel('Month')
plt.ylabel('Total Revenue')
plt.grid(True)
plt.show()
```



[14]:
```
# Sales Trend by order priority using line chart.
priority_monthly_sales = data.groupby(['Month', 'Order Priority'])['Total␣
↪Revenue'].sum().reset_index()

# Convert 'Month' from Period to string for plotting
priority_monthly_sales['Month'] = priority_monthly_sales['Month'].astype(str)

# Plot the sales trends by order priority
plt.figure(figsize=(12, 6))
sns.lineplot(x='Month', y='Total Revenue', hue='Order Priority',␣
↪data=priority_monthly_sales, marker='o')
plt.xticks(rotation=45)
plt.title('Sales Trends by Order Priority')
plt.xlabel('Month')
plt.ylabel('Total Revenue')
plt.legend(title='Order Priority')
```

```
plt.grid(True)
plt.show()
```